SmartIndicator

Complete English Documentation

Elmer Petry

Version 1.5.1

Updated: August 22, 2026

Contents

1 Introduction

1.1 Overview of SmartIndicator

In astronomical images, indicators are used to visually represent specific information. SmartIndicator is a PixInsight script that allows you to overlay different types of such indicators on astronomical images. Supported indicators include:

- **Orientation Indicator** → Shows the orientation (rotation) of the image.
- **Image Scale Indicator** → Shows the image scale.
- **Angular Separation Indicator** → Shows the angular distance between two points.

Examples:

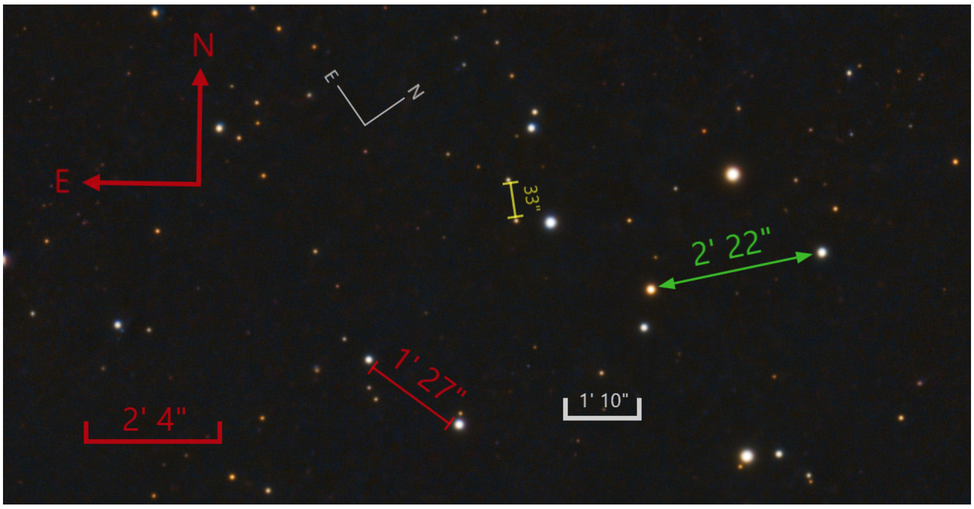


Figure 1: Indicators

The size, position, color, and transparency of the indicators are fully adjustable. Indicators can be drawn directly onto the target image, applied to a copy, or generated as overlays — either complete or shape-only. All changes to the target image are reversible (Undo/Redo) and affect only the pixels within the indicator region.
A key idea behind SmartIndicator is to enable repeatable, consistent placement of indicators across different images. For this reason, the size and position of indicators are stored relative to the image rather than as absolute values. SmartIndicator also allows you to create *Process Icons* for semi-automated application of indicators.
For example, to place the same indicator at the same relative position in multiple images, you configure it once and store it as a Process Icon in the PixInsight workspace. Process Icons such as “NE-indicator-red-top-left” or “Scale-gray-bottom-right” can be created. These icons can then be applied to different images simply by dragging them onto the image. The indicators will always appear at the same relative position and size.

1.2 Underlying Principle

The only requirement for using SmartIndicator is an image with a valid astrometric solution. SmartIndicator uses the astrometric solution to:

- determine the rotation and orientation of the image for the Orientation Indicator,
- calculate the correct length of the Image Scale Indicator based on the angular resolution per pixel,
- measure angular distances in arcseconds.

The basic workflow for adding an indicator is intuitive and simple:

1. Open the target image.
2. Start SmartIndicator.
3. Place the indicator.
4. Apply.

2 Installation

2.1 Adding the Repository

To install SmartIndicator in PixInsight (PI), the script repository must be added. Open `RESOURCES → Updates → Manage Repositories` in the main menu and select `Add`. Then enter the following URL completely — including the trailing slash:

```
https://raw.githubusercontent.com/ElmerPetry/pi-scripts/updates/
```

PI stores repository entries permanently, so future updates are detected automatically.

2.2 Running the Installation

After adding the repository, SmartIndicator will be installed automatically the next time PI is started. Alternatively, the installation can be triggered immediately via `RESOURCES → Updates → Check for Updates`.
During installation, PI downloads all new scripts and modules from the repository and integrates them into the existing environment.

2.3 Verifying the Installation

After successful installation, SmartIndicator is available in the script menu:

`Script → Render → SmartIndicator`

On first launch, the user interface is initialized and the default parameters are loaded.

2.4 Notes and Troubleshooting

If the installation is not successful, double-check the following prerequisites hold:

- correct repository URL (especially the trailing slash),
- working internet connection,
- proxy or firewall settings,
- sufficient permissions for PI.

Running `Check for Updates` again resolves many temporary issues.

3 Usage

3.1 Starting the Script

The script is launched through the following PixInsight menu:

Script → Render → SmartIndicator

The main window will then appear.

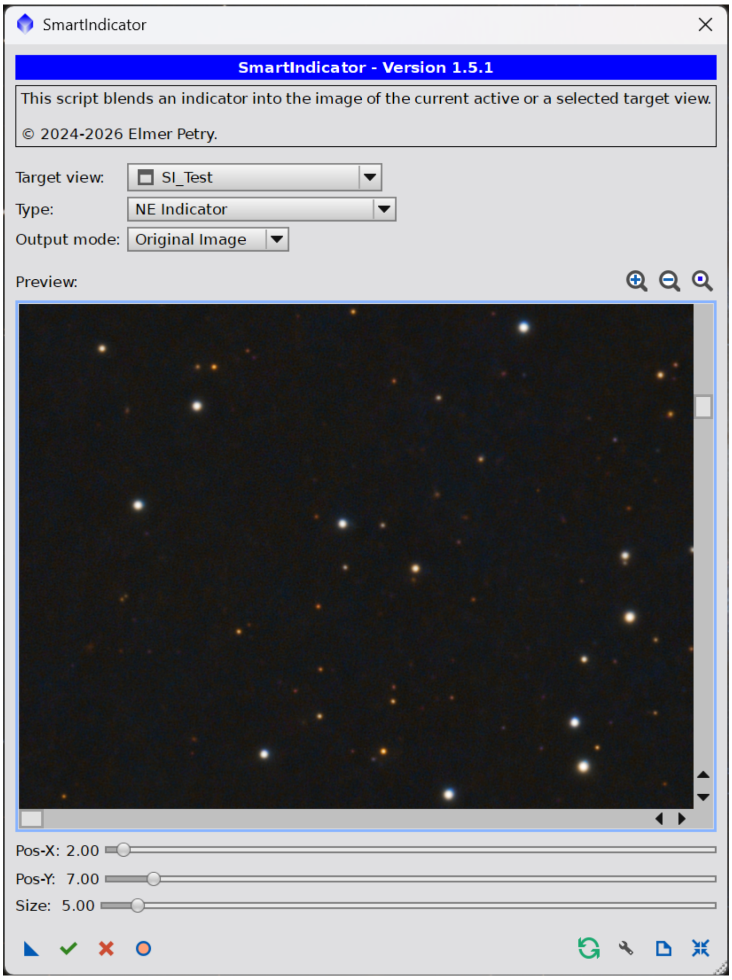


Figure 2: Main Window

3.2 Main Window

The main window is intuitive to use and stores settings permanently (see Figure 2). By dragging the lower-right corner, the entire main window can be enlarged or reduced.

3.2.1 Global Options

Target view: Specifies the target image.

The active image in the workspace is automatically preselected as the target image. A valid astrometric solution is required. If the target image does not have an astrometric solution, a corresponding message will be displayed in the console and no indicator can be generated.

Type: Determines the type of indicator.

The following indicators are available:

- **NE Indicator**
- **Image Scale Indicator**
- **Angular Separation Indicator**

Output mode: Defines the output type.

The following output types are available:

- **Original Image** → The indicator will be drawn directly onto the target image.
- **Copy of Image** → A copy of the target image with the indicator will be created.
- **Overlay** → A transparent overlay containing the indicator will be generated.
- **Shape** → A transparent shape containing only the indicator geometry will be generated.

3.2.2 Preview

Displays a preview area of the image.
The preview region can be selected using the sliders on the sides. The preview can also be scaled (zoom function) — either by turning the middle mouse wheel or by clicking the following icons above the preview area:

Increases the zoom level.

Decreases the zoom level.

Resets the zoom to a 1:1 view.

The maximum zoom level corresponds to 400% of the image size, which is fully sufficient for precise placement. The minimum zoom level is reached when the entire image is visible in the preview area.
The indicator is placed in the preview area using the left mouse button and can be freely positioned by dragging while holding the button. If the indicator touches an image edge, it automatically snaps so that it remains fully visible and does not extend beyond the image boundary.
All changes are shown immediately in real time and can be adjusted repeatedly as desired. Once the preview results fully matches the expectations it might be applied to the target image (see Section 3.2.4).

3.2.3 Indicator-specific Options

Below the preview area, additional settings appear depending on the selected indicator. Further properties such as color, transparency, font size, etc. can also be adjusted. These options are described in the chapters for each indicator.

3.2.4 Toolbar

At the bottom of the main window, a toolbar with icons for various functions is displayed. Each function is triggered simply by clicking the corresponding icon. The functions from left to right are:

- Creates a new Process Icon by dragging it onto the workspace.
- Apply changes.
- Cancel.
- Toggle preview on/off.
- Refresh preview.
- Open settings dialog.
- Show documentation.
- Reset all settings to default values.

3.2.5 Notes

Position: Indicators are automatically adjusted if they would extend beyond the image boundary.

Size: The size is also automatically corrected if necessary. Line width, font size, or arrowhead dimensions are scaled down if they would otherwise be too large relative to the indicator.

Console: The console displays additional information such as rotation, resolution, and indicator dimensions. It also shows hints and — if necessary — warnings or error messages that point to the respective issue.

3.3 Orientation Indicator

3.3.1 Purpose and Functionality

The Orientation Indicator (NE Indicator) is used to display the orientation of an image. Typically, this is shown by indicating the directions of North (N) and East (E) within the image.

Tip:

> If East is not displayed to the left of North, the image is mirrored. In this case, it is recommended to correct the image by mirroring it. Thus, the indicator can also be used simply to verify the correct orientation of an image.

3.3.2 Placement of the Indicator

The Orientation Indicator is placed by dragging it with the left mouse button inside the preview area.

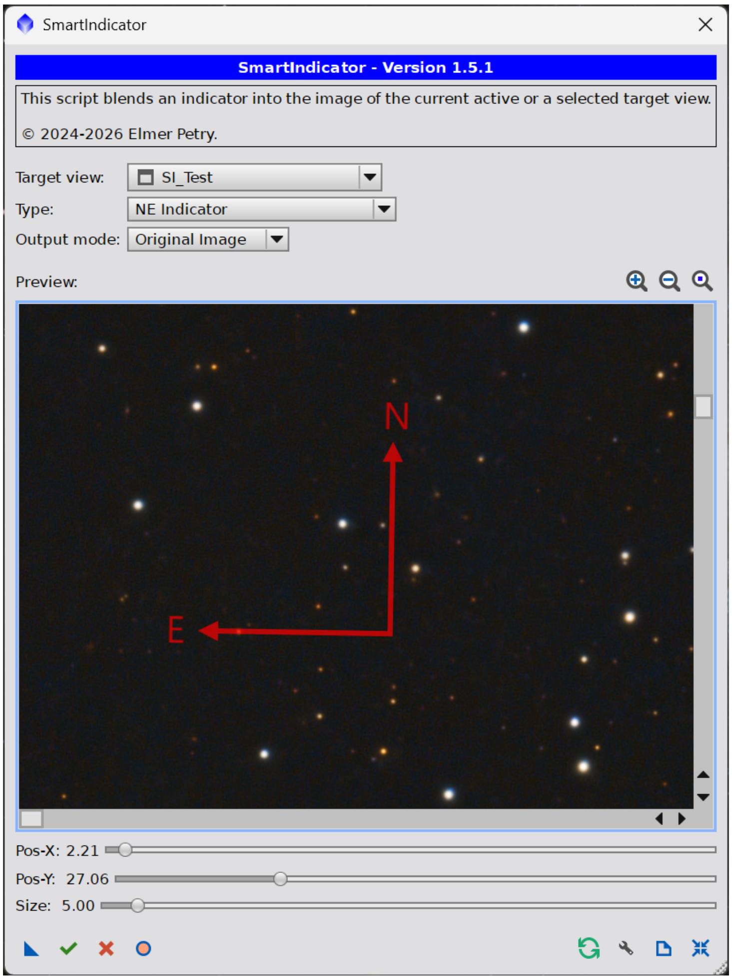


Figure 3: Main window with Orientation Indicator

Below the preview area, the following settings are available:

Pos-X: / Pos-Y: Change the position of the indicator.

These sliders adjust the position of the indicator as a percentage of the image width or height. When a slider is selected, it can also be adjusted using the keys *Page Up* or *Page Down* in steps, or with ↑ and ↓ in hundredth-steps for precise adjustments.

Size: Change the size of the indicator.

A slider to adjust the size of the Orientation Indicator relative to the image dimensions.

3.3.3 Settings Dialog

You can access the settings dialog via the icon in the toolbar.

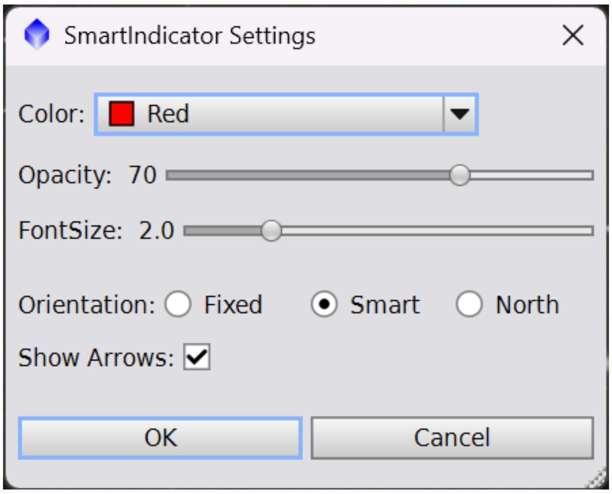


Figure 4: Settings dialog for the Orientation Indicator

The following properties can be adjusted:

Color: Color of the indicator.

Opacity: Transparency level (in percent).

FontSize: Font size of the direction letters (as a percentage of the image height).

Orientation: Orientation of the direction letters (N and E).

- **Fixed:** Orientation follows the rotation of the indicator.
- **Smart:** Orientation follows the rotation of the indicator, but for certain rotation angles it is adjusted to match the reading direction.
- **North:** Always upright, regardless of indicator rotation.

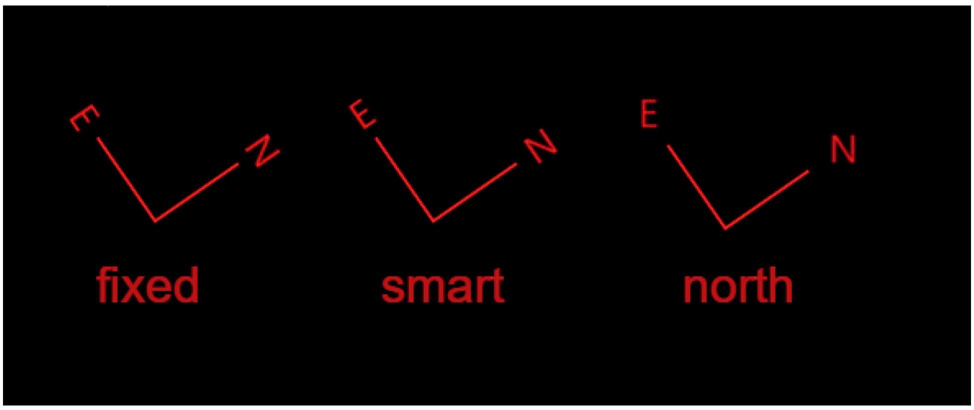


Figure 5: Examples of orientation at a rotation of 55.4°

Show Arrows: Determines whether arrowheads should be displayed at the ends.

3.3.4 Console Output

The console displays the following information for the Orientation Indicator:

Rotation: Rotation angle of the indicator and thus of the image.

Size: Side length of the indicator in pixels and the equivalent in arcseconds.

PosX: X-coordinate of the indicator's bottom-left point in the image pixel grid.

PosY: Y-coordinate of the indicator's bottom-left point in the image pixel grid.

```
NE Indicator:
Rotation: 0.918 deg
Size:     153 px = 106.335 arcsec
PosX:     152
PosY:     4172
```

Figure 6: Console output for the Orientation Indicator

3.4 Image Scale Indicator

3.4.1 Purpose and Functionality

The Image Scale Indicator shows the scale of the image. Every image has an astrometrically determinable resolution in arcseconds per pixel. The indicator displays the angular distance corresponding to this resolution between the endpoints of a line.

3.4.2 Placement of the Indicator

The Image Scale Indicator is placed by dragging with the left mouse button inside the preview area.

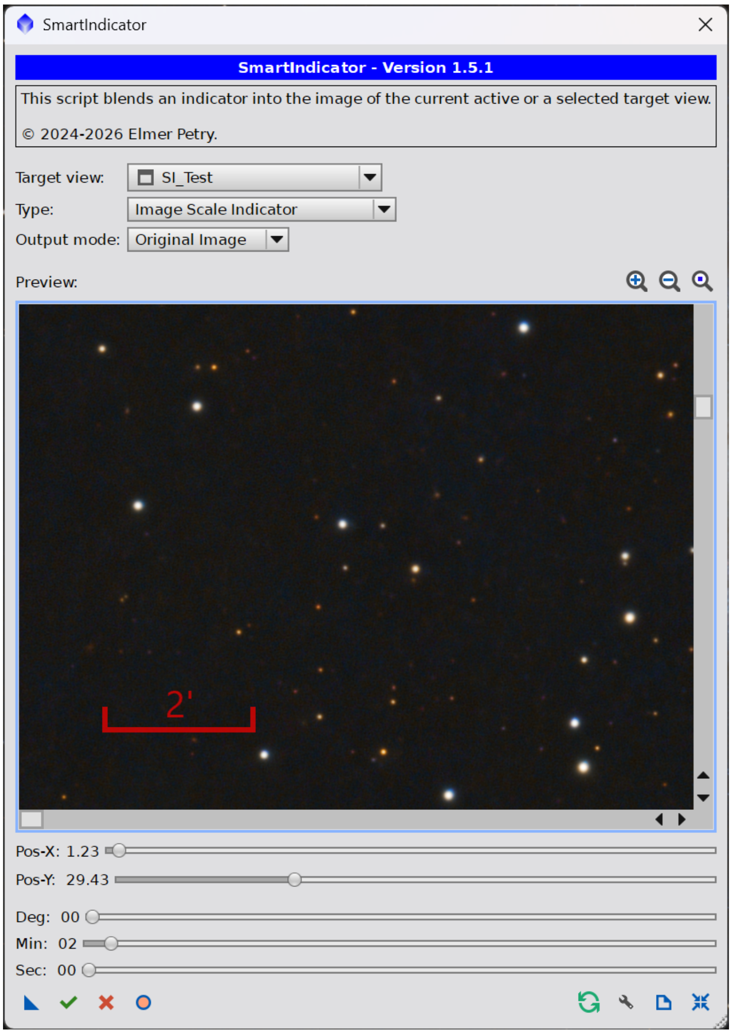


Figure 7: Main window with Image Scale Indicator

Below the preview area, the following settings are available:

Pos-X: / Pos-Y: Change the position of the indicator.

These sliders adjust the position of the indicator as a percentage of the image width or height. When a slider is selected, it can also be adjusted using the keys *Page Up* or *Page Down* in steps, or with ↑ and ↓ in hundredth-steps for precise adjustments.

Deg: / Min: / Sec: Specify the angle to be displayed.

These sliders allow you to set the angle that the indicator should represent, in degrees, arcminutes, and arcseconds. When a slider is selected, it can also be adjusted using the keys *Page Up* or *Page Down* in 5-step increments, or with ↑ and ↓ in single-step increments.

Note that the maximum length is limited by the image size and image scale, and SmartIndicator automatically accounts for this. For example, if an image spans 1 degree and 20 arcminutes across its width, and the sliders are set to 3 degrees, SmartIndicator will automatically reduce the values to the maximum possible length (in this example, 1 degree and 20 arcminutes).

Tip:

> The automatic adjustment for size limits can be used to determine the total angular width of an image. Simply drag the degree slider to the right and release it. The slider values, the indicator, and the console output will immediately show the angular width of the entire image.

3.4.3 Settings Dialog

You can access the settings dialog via the icon in the toolbar.

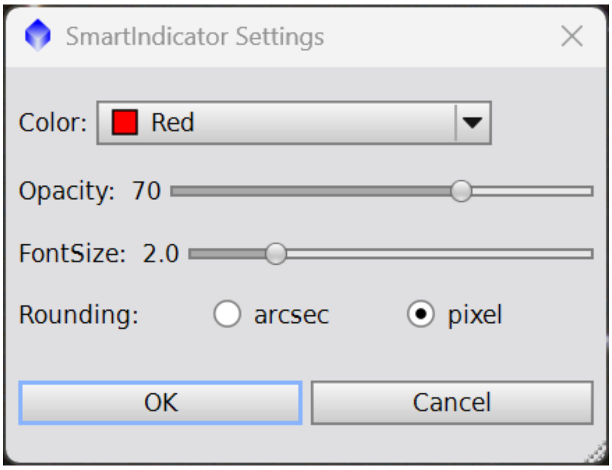


Figure 8: Settings dialog for the Image Scale Indicator

The following properties can be adjusted:

Color: Color of the indicator.

Opacity: Transparency level (in percent).

FontSize: Font size of the text (as a percentage of the image height).

Rounding: Type of rounding used when calculating the length.

- **arcsec:** The angle displayed by the indicator is rounded to whole arcseconds, specifically to the value whose corresponding line length is closest to the actual length of the indicator. This is the most accurate rounding method.
- **pixel:** The specified angle in arcseconds is displayed without modification, and the indicator is drawn to the pixel whose position is closest to this arcsecond value. In most cases this is less precise, but the displayed angle can be specified exactly, which may result in a more visually appealing presentation.

3.4.4 Console Output

The console displays the following information for the Image Scale Indicator:

Setting: Selected length of the indicator in arcseconds.

Resolution: Resolution in arcseconds per pixel.

Size: Length of the indicator in pixels, the equivalent in arcseconds, and the values in degrees + arcminutes + arcseconds.

Deviation: Accuracy or deviation in percent.

PosX: X-coordinate of the indicator's bottom-left point in the image pixel grid.

PosY: Y-coordinate of the indicator's bottom-left point in the image pixel grid.

```
Image Scale Indicator:
Setting:    121 arcsec
Resolution: 0.695 arcsec/px
Size:       174 px = 121 arcsec ≈ 2' 1"
Deviation:  0.06%
PosX:       648
PosY:       514
```

Figure 9: Console output for the Image Scale Indicator

3.5 Angular Separation Indicator

3.5.1 Purpose and Functionality

The Angular Separation Indicator displays the angular distance between two arbitrary points in the image. It is typically used to indicate the angular separation of two astronomical objects or the size or extent of objects in the image.

3.5.2 Placement of the Indicator

The indicator is placed in the preview area simply by dragging with the left mouse button from the starting point to the endpoint. During this process, the text is automatically aligned and positioned according to the indicator. After the indicator has been set, the endpoints can be individually moved and finely adjusted with the right mouse button if needed. The text can also be moved independently with the right mouse button. This can be useful, for example, in cases where the text would otherwise accidentally cover an important area of the image.

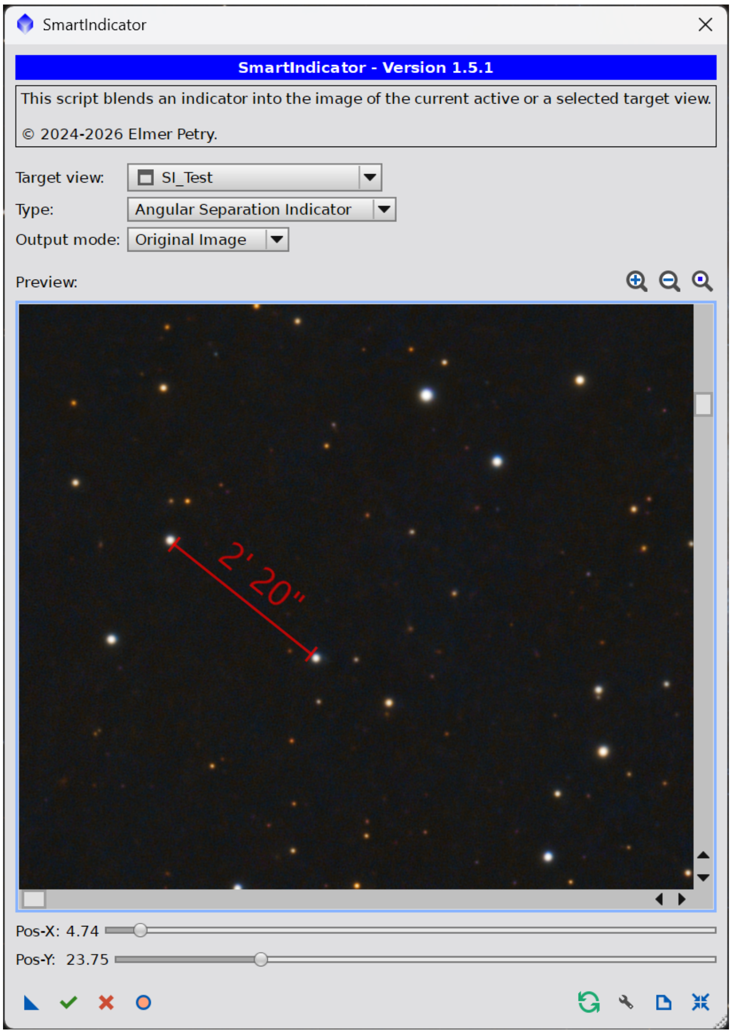


Figure 10: Main window with Angular Separation Indicator

Below the preview area, the following settings are available:

Pos-X: / Pos-Y: Change the position of the indicator.

These sliders adjust the position of the indicator as a percentage of the image width or height. When a slider is selected, it can also be adjusted using the keys *Page Up* or *Page Down* in steps, or with ↑ and ↓ in hundredth-steps for precise adjustments.

3.5.3 Settings Dialog

You can access the settings dialog via the 🔧 icon in the toolbar.

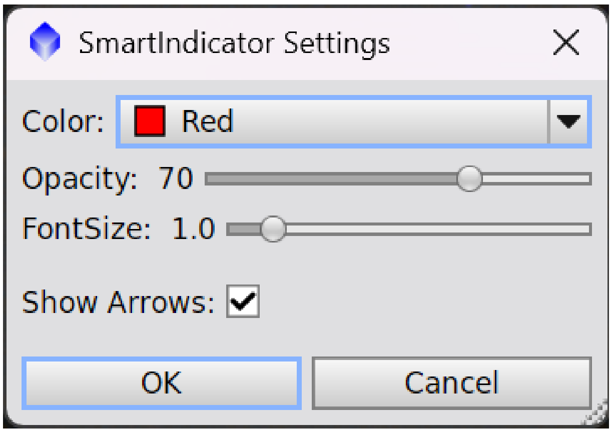


Figure 11: Settings dialog for the Angular Separation Indicator

The following properties can be adjusted:

Color: Color of the indicator.

Opacity: Transparency level (in percent).

FontSize: Font size of text (as a percentage of the image height).

Show Arrows: Determines whether arrowheads should be displayed at the ends. If not, end markers will be shown.

3.5.4 Console Output

The console displays the following information for the Angular Separation Indicator:

Setting: Length of the indicator in arcseconds.

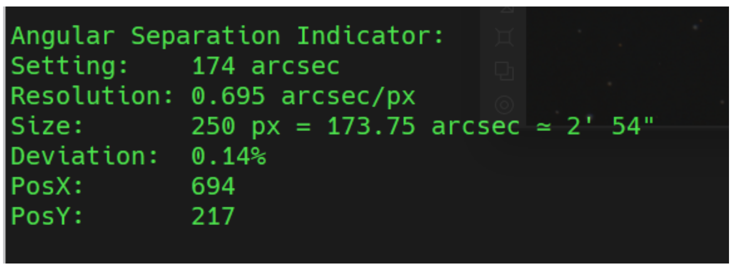
Resolution: Resolution in arcseconds per pixel.

Size: Length of the indicator in pixels, the equivalent in arcseconds, and the values in degrees + arcminutes + arcseconds.

Deviation: Accuracy or deviation in percent.

PosX: X-coordinate of the indicator's bottom-left point in the image pixel grid.

PosY: Y-coordinate of the indicator's bottom-left point in the image pixel grid.



```
Angular Separation Indicator:
Setting:     174 arcsec
Resolution:  0.695 arcsec/px
Size:        250 px = 173.75 arcsec ≈ 2' 54"
Deviation:   0.14%
PosX:        694
PosY:        217
```

Figure 12: Console output for the Angular Separation Indicator

4 Process Icons

4.1 Creation and Application

By dragging the icon from the toolbar (see Section 3.2.4) onto the workspace, a so-called *Process Icon* can be created. A Process Icon stores all current settings of an indicator. It is possible to assign a custom name to a Process Icon. To do this, select the Process Icon and choose the menu item *Set Icon Identifier...* from the popup menu.

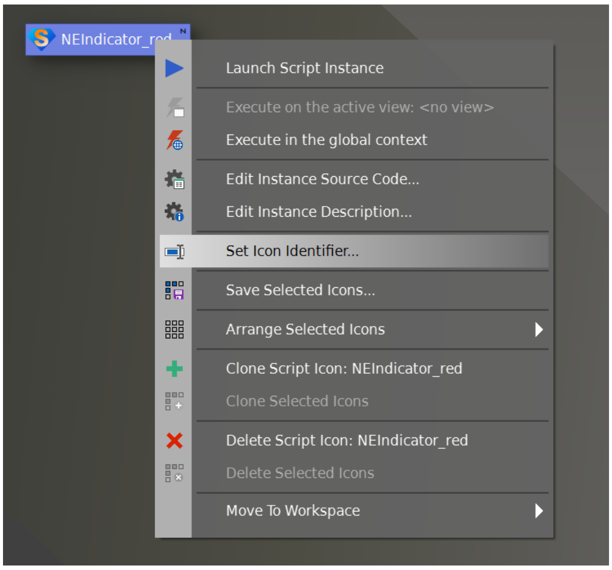


Figure 13: Renaming a Process Icon

For different indicators in different configurations, any number of Process Icons can be created. These Process Icons can be saved separately and later loaded into workspaces. Since all relevant indicator parameters are stored by SmartIndicator as relative values rather than absolute ones, the Process Icons can be applied to different images — even those with different dimensions and resolutions — without any issues.
Applying a Process Icon to an image is done simply by dragging the icon onto the image. This allows indicators in predefined configurations to be applied almost automatically. For example, if you always want an Orientation Indicator in the top-left corner and an Image Scale Indicator in the bottom-right corner, you configure each indicator once and create the corresponding Process Icons. One might be named “NE-top-left” and the other “Scale-bottom-right”. After that, you simply drag these Process Icons onto all images that should contain these indicators.
The indicators will then appear in all images at the same relative position and size — and of course with all other properties such as color, transparency, etc. If an image exists in different resolutions, the indicator will still appear consistently relative to the image dimensions.

Note on Updates:

When updating SmartIndicator to a newer version, previously saved process icons may become incompatible, resulting in a warning message. This is due to a hash code check performed before execution to ensure that the script has not changed since the icon was created. In most cases, updates remain backward-compatible. If so, you can copy the hash code from a process icon generated with the new version and continue using your existing icon.

4.2 Common Parameters

The following parameters are stored in every process icon and can be manually adjusted if needed:

targetView
: The target image to which the indicator will be assigned.

outputMode

- **0** Original Image – Draws directly onto the target image.
- **1** Copy of Image – Creates an annotated copy of the target image.
- **2** Overlay – Creates a transparent overlay with the same dimensions and astrometric solution as the target image.
- **3** Shape – Creates a small transparent overlay in the shape of the indicator.

fontSizePercent
: Font size as a percentage of the image height.

color
: Color of the indicator.

indicatorOpacityPercent
: Opacity of the indicator in percent.

indicatorType

- **0** NE Indicator
- **1** Image Scale Indicator
- **2** Angular Separation Indicator

4.3 Parameters of the NE Indicator

neIndicatorPosXpercent
: Horizontal position of the indicator as a percentage of the image width.

neIndicatorPosYpercent
: Vertical position of the indicator as a percentage of the image height.

neIndicatorSize
: Size of the indicator as a percentage relative to the image.

neIndicatorOrientationMode

- **0** Fix – The letters N and E are fixed relative to the indicator.
- **1** Smart – Same as Fix, but in certain rotation angles the letters are rotated to match the reading direction (default).
- **2** North – N and E are always oriented vertically.

neIndicatorArrows

- **true** show arrowheads (default).
- **false** do not show arrowheads.

4.4 Parameters of the Image Scale Indicator

indicatorArcSec
Length of the indicator in arcseconds.

indicatorPosXpercent
Horizontal position as a percentage of the image width.

indicatorPosYpercent
Vertical position as a percentage of the image height.

indicatorRoundingArcSec

- **true** The angle displayed by the indicator is rounded to whole arcseconds, specifically to the value whose corresponding line length is closest to the actual length of the indicator. This is the most accurate rounding method.
- **false** The specified angle in arcseconds is displayed without modification, and the indicator is drawn to the pixel whose position is closest to this arcsecond value. In most cases this is less precise, but the displayed angle can be specified exactly, which may result in a more visually appealing presentation.

4.5 Parameters of the Angular Separation Indicator

asIndicatorStartPosXpercent
Horizontal position as a percentage of the image width of the indicator's starting point.

asIndicatorStartPosYpercent
Vertical position as a percentage of the image height of the indicator's starting point.

asIndicatorTextPosXpercent
Horizontal position as a percentage of the image width of the text position.

asIndicatorTextPosYpercent
Vertical position as a percentage of the image height of the text position.

asIndicatorArcSec
Length of the indicator in arcseconds based on the image scale.

asIndicatorDirectionAngle
Direction angle of the indicator.

asIndicatorArrows

true show arrowheads (default).

false show orthogonal end marks.

5 Technical Fundamentals

5.1 Pixel Grid

Every digital image has a discrete pixel grid and a specific angular resolution per pixel. The angular resolution per pixel is usually given in arcseconds per pixel. For astronomical images, this resolution can be accurately determined by matching the image against star catalogs (Lang et al. 2010). The angular resolution and additional information of astrometrically solved images are stored by PixInsight in the image metadata and are read and used by SmartIndicator.

5.2 Computation of Lengths from Angles

In an image with an angular resolution per pixel of θ_{px}, the length L of a line corresponding to a given angular separation θ is generally computed as follows:

$$L = \frac{\theta}{\theta_{\text{px}}}$$

A challenge in computing the length or in displaying it in the form of a scale indicator arises from the discrete pixel grid of an image. The length of the indicator cannot be displayed continuously, but only in whole pixels:

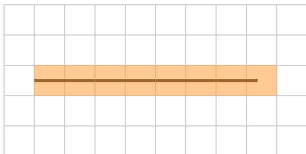


Figure 14: Illustration of pixel-grid discretization

To illustrate this situation, consider the following example:
Assume an image with an angular resolution of 1.789 arcseconds per pixel. For a target angle of 3 arcminutes, the ideal length in pixels is

$$N_{\text{px}} = \frac{3'}{\theta_{\text{px}}} \Rightarrow N_{\text{px}} = \frac{3 \cdot 60''}{1.789''/\text{px}} \approx 100.615 \text{ px}$$

Due to the discrete pixel grid, however, the indicator can only be displayed using whole pixel values, i.e., with 100 px or 101 px. For the total angle over N pixels, the following holds:

$$\theta(N_{\text{px}}) = N_{\text{px}} \cdot \theta_{\text{px}}$$

Now insert the values from our example:

$$\theta_{\text{px}} = 1.789''/\text{px}, \quad N_{\text{px}} = 101\text{px} \quad \Rightarrow \quad \theta(101) = 101\text{px} \cdot 1.789''/\text{px} = 180.689'',$$

which corresponds to a deviation of

$$\Delta\theta = \theta(101) - 180'' = 0.689''$$

$$\Rightarrow \quad \frac{\Delta\theta}{180''} \cdot 100 = \frac{0.689''}{180''} \cdot 100 \approx 0.38\%$$

from the nominal value.

5.3 Rounding Modes (arcsec / pixel)

SmartIndicator offers two variants for computing and displaying the indicators. The user can therefore decide whether accuracy or adherence to the specified angle is preferred. This can be configured in the settings under *Rounding* (see Figure 8). If *pixel* is selected, then — as in the previous example — the indicator for 3 arcminutes would be displayed with a length of 101 pixels.
If, on the other hand, *arcsec* is chosen as the rounding mode, the indicator would also have a length of 101 pixels in this example, but SmartIndicator would adjust the angle specification by one arcsecond to 3 arcminutes and 1 arcsecond. In this case, the deviation becomes:

$$\theta_{\text{px}} = 1.789''/\text{px}$$

$$\theta(101) = 101\text{px} \cdot 1.789''/\text{px} = 180.689''$$

$$\Delta\theta = \theta(101) - 181'' = -0.311''$$

$$\Rightarrow \quad \frac{\Delta\theta}{181''} \cdot 100 = \frac{-0.311''}{181''} \cdot 100 \approx -0.17\%$$

Many amateurs work with resolutions in the range of $0.8 - 2.0''/\text{px}$. The deviation naturally depends on the resolution and the length of the indicator. In the example, the deviations for an indicator corresponding to one arcminute would be 1.38% for *pixel* and 0.29% for *arcsec*. The following table shows the results of both rounding methods for $\theta_{\text{px}} = 1.789''/\text{px}$ across several predefined angular distances:

Angle	**Ideal (px)**	**Indicator (px)**	**arcsec**	**pixel**
$1'$	33.54	34	$1'1''$ (0.29%)	$1'$ (1.38%)
$2'$	67.08	67	$2'$ (0.11%)	$2'$ (0.11%)
$3'$	100.62	101	$3'1''$ (0.17%)	$3'$ (0.38%)
$5'$	167.67	168	$5'1''$ (0.15%)	$5'$ (0.18%)
$10'$	335.38	335	$9'59''$ (0.05%)	$10'$ (0.11%)

Beyond a certain indicator length, the differences in accuracy become negligible for most practical purposes. To allow verification of the respective deviation when needed, it is displayed in the console (see Figure 9).

Recommendation:

For scientifically accurate values, *arcsec* is recommended; for visual consistency, *pixel*.

References

Lang, Dustin et al. (2010). "Astrometry.net: Blind astrometric calibration of arbitrary astronomical images". In: *The Astronomical Journal* 139.5, pp. 1782–1800. DOI: 10.1088/0004-6256/139/5/1782.

6 Disclaimer and license information

The SmartIndicator script is based on software from the PixInsight project, developed by Pleiades Astrophoto and its contributors (`https://pixinsight.com/`).